

Backend Developer Assignment

Objective

Build a REST API application using Node.js, TypeScript, Express.js, MongoDB and Mongoose.

Mandatory Tech Stack

Node.js, TypeScript, Express.js, MongoDB, Mongoose.

Scenario

Implement a backend for a platform where users can book sessions with teachers. Teachers create available sessions, users can book them, and completed sessions should appear in the user's history.

Database Schemas

User

fullName, email (unique), phone, timestamps.

Teacher

fullName, email, specialization, experience, timestamps.

Session

teacherId (ObjectId), userId (ObjectId | null), startTime, endTime, status (AVAILABLE / BOOKED / COMPLETED), completedAt, timestamps.

API 1 - Create User

POST /users

Create a new user. Email must be unique.

API 2 - Create Session

POST /sessions

Create a new session for a teacher. Default status should be AVAILABLE. Validate teacher exists and endTime is greater than startTime.

API 3 - Available Sessions for Booking

GET /sessions/available?dateTimestamp={timestamp}

Return all AVAILABLE sessions for the supplied date.

Mandatory: This API must be implemented using the MongoDB Aggregation Pipeline.

API 4 - Book Session

POST /sessions/:id/book

Rules:

- User must exist.
- Session must exist.
- Only AVAILABLE sessions can be booked.
- A booked session cannot be booked again.
- Update session status to BOOKED and associate it with the user.

API 5 - Mark Session Complete

PATCH /sessions/:id/complete

Rules:

- Session must exist.
- Only BOOKED sessions can be marked as completed.
- Update status to COMPLETED and set completedAt.

API 6 - User Session List

GET /users/:id/sessions

Return all sessions for the user separated into:

- Upcoming Sessions
- Completed Sessions

Mandatory: This API must be implemented using the MongoDB Aggregation Pipeline.

General Requirements

- Use proper HTTP status codes.
- Centralized error handling.
- Async/await only.
- Environment variables for configuration.
- Follow clean coding practices.
- Provide a README with setup instructions.
- Submit a Postman collection for API testing.